

Private Pilot (ASEL) Oral Exam Question Bank

A knowledge-focused study resource for the Private Pilot Airplane Single-Engine Land checkride oral exam. Organized by topic area, with a complete model answer for every question.

Two tips before you start studying: - **Answer the question asked, then stop.** Volunteering extra information beyond what was asked often invites a harder follow-up question that wasn't coming. If the examiner wants more detail, they'll ask for it. - **Speak in your own words, not acronyms.** An examiner can tell when an answer is recited from a mnemonic rather than understood — be ready to explain the substance behind any acronym you use.

1. Certificates, Privileges, and Currency

1. What documents must you have in your possession to act as pilot-in-command? **A:** Government-issued photo ID, pilot certificate, and current medical certificate (unless operating under BasicMed, or as a glider/balloon pilot exempt from the requirement).
2. What are the privileges and limitations of your certificate? **A:** You may act as PIC of an aircraft in the category/class/type on your certificate for any purpose not for compensation or hire, carry passengers and property, and share operating expenses on a pro-rata basis. Narrow exceptions let you be compensated: PIC of a charitable/nonprofit/community-event flight under specific conditions, search-and-rescue operations for a nonprofit, or acting as an aircraft salesperson with at least 200 hours logged. Outside those exceptions, you may not be paid for acting as PIC or carry passengers/property for compensation or hire.
3. You own a business — can you fly your own plane on the clock to go look at something for it, such as scouting a vehicle in another state? **A:** Yes. As long as you aren't being paid specifically for the fact that you're flying — the flight has to be "incidental to the business." You could have driven or taken a commercial flight instead; the business purpose is what's being compensated, not the piloting.
4. A friend offers to pay you to fly him somewhere to pick up a car he bought. Can you accept payment for that flight? **A:** No — that's compensation for the flight itself, which a private pilot can't accept, unless you had your own independent reason ("common purpose") to make that same trip anyway, in which case you can share the pro-rata operating expenses (fuel, oil, aircraft rental, landing/tie-down fees — not the whole cost, and never more than an equal share for the pilot) with him.
5. Can you post on social media asking if anyone wants to fly with you somewhere and split gas? **A:** No — that's "holding out," publicly offering your flight to whoever wants it, which starts to look like common carriage. Sharing pro-rata expenses is only legal with someone you already had a reason to invite, not an open solicitation.
6. What's required for you to legally carry passengers, and what's the "gotcha" about the different definitions of night? **A:** Three takeoffs and three landings to a full stop in the same category/class (and type, if required) of aircraft within the preceding 90 days. If any of that passenger-carrying will be at night, those landings must also be to a full stop (touch-and-goes don't count) and made in the window from one hour after sunset to one hour before sunrise — meeting the night requirement also satisfies day currency.

Aviation actually uses three different, non-aligned definitions of "night": sunset-to-sunrise is when position/anti-collision lights are required (91.209); end of evening civil twilight to beginning of morning civil twilight is what you log as night flight time (61.51); and one-hour-after-sunset to one-hour-before-sunrise is the window for passenger-carrying currency (61.57(b)). Examiners like to test whether you know which definition governs which requirement.

7. What's the difference between currency and proficiency? **A:** Currency is the minimum legal requirement (e.g., 3 landings in 90 days) to exercise a privilege. Proficiency is whether you're actually sharp enough to do it safely — you can be legally current and still not proficient, especially after a long layoff, and personal minimums should account for that gap.

8. How do you keep your certificate legally current to act as PIC at all? **A:** Complete a flight review — 1 hour of ground training and 1 hour of flight training with an authorized instructor — every 24 calendar months, with a logbook endorsement. Certain things substitute for it: earning a new certificate or rating, a proficiency check, or completing a phase of the FAA WINGS program.

9. What are the classes of medical certificate, how long is each valid, and how do the higher classes downgrade over time? **A:** First class (ATP privileges), second class (commercial privileges), and third class (private/recreational/student privileges) — a higher class also satisfies any lower class while its own privileges remain valid. Validity depends on age at the time of the exam. Under age 40: a third-class medical is valid 60 calendar months; a first- or second-class medical carries its higher-class privileges for 12 months, then automatically downgrades to third-class privileges through the end of the 60th month. Age 40 or older: a third-class medical is valid 24 months; a first-class medical gives first-class privileges for 6 months, then second-class privileges for the next 6, then third-class through month 24; a second-class medical gives second-class privileges for 12 months, then third-class through month 24. A medical always expires on the last day of its specified expiration month.

10. What is BasicMed, and what are its requirements and limitations? **A:** An alternative to holding an FAA medical certificate for certain flights. Requirements: you held a valid FAA medical certificate at some point after July 14, 2006; your most recent medical application wasn't denied, revoked, or suspended; you complete a comprehensive physical exam with a state-licensed physician every 48 months using the FAA's checklist; and you complete a free FAA-approved online medical education course every 24 months, carrying proof of completion. Limitations: aircraft authorized for takeoff with 6 or fewer occupants and a max certificated takeoff weight of 12,500 lbs or less; VFR or IFR within the US at or below 18,000 ft MSL, no faster than 250 KIAS; no more than 6 passengers; not for compensation or hire.

11. What are the alcohol and drug rules for pilots? **A:** You may not act as a crewmember within 8 hours of consuming alcohol ("8 hours bottle to throttle"), while under the influence, with a blood alcohol concentration of 0.04 or greater, or while using any drug that affects your faculties in a way contrary to safety. You also may not allow anyone who appears intoxicated or impaired aboard, except a medical patient under proper care.

12. What is IMSAFE, and when do you use it? **A:** A personal self-assessment checklist run before every flight: Illness, Medication, Stress, Alcohol, Fatigue, Eating/hydration. If any item is unsatisfactory, don't fly. It's most useful compared against your own personal minimums, which should be more conservative than the legal minimums.

13. What must you do if you develop a medical condition that no longer meets your medical certificate's standards? **A:** You may not act as PIC (or any required crewmember) while you know, or have reason to know, of a medical deficiency, or while taking medication that makes you unable to meet the certificate's standards. Self-grounding is the pilot's own responsibility — having a valid medical certificate in hand doesn't override this.
14. What happens if you fail a portion of the practical test? **A:** The examiner issues a Notice of Disapproval listing the failed Areas of Operation. You may request a partial retest of just those items — typically within 60 days, once retrained and endorsed by an instructor. If more than 60 days pass, a full retest is required.

2. Aircraft Documents & Airworthiness

15. What documents must be in the aircraft for it to be legal to fly? **A:** Airworthiness certificate, Registration, Radio station license (only required for international operations), Operating limitations/handbook (POH/AFM), and Weight and balance data — plus any equipment-specific supplements (GPS, autopilot, engine monitor, etc.) that came with installed avionics. (Common mnemonic: ARROW.) The airworthiness certificate must be displayed conspicuously — typically the cabin or cockpit entrance — so it's legible to passengers and crew.
16. Does the airworthiness certificate expire? **A:** No — it remains valid as long as the aircraft meets its type design and is maintained in accordance with 14 CFR. The registration certificate, by contrast, is valid for 7 years from issuance (or until sold, revoked, or the aircraft is destroyed) and is non-transferable.
17. What is "airworthy," and who makes that determination? **A:** The aircraft conforms to its type certificate (or approved alterations) and is in condition for safe operation. The pilot in command makes the final call before every flight, based on required inspections, ADs, and a satisfactory preflight.
18. What inspections does the aircraft require, and how often? **A:** Annual — every 12 calendar months, by an A&P with Inspection Authorization. 100-hour — every 100 hours of operation, but only required if the aircraft is used for hire or flight instruction for hire (an annual inspection satisfies the 100-hour requirement, but not the reverse). ADs — Airworthiness Directives, complied with on whatever schedule each one specifies (one-time, recurring, or by a compliance date). VOR check — every 30 days, only if flown IFR using VOR navigation. Transponder — every 24 calendar months. ELT — every 12 calendar months, and also after 1 cumulative hour of use or when 50% of its battery life has expired. Static system/altimeter/pitot — every 24 calendar months, required only for IFR flight. (Mnemonic: AV1ATES.)
19. Can an overdue annual or 100-hour inspection be flown over? **A:** An overdue annual grounds the aircraft outright — it can only be flown under a special flight permit (ferry permit) from the FSDO, typically to get it to a shop. A 100-hour inspection may be exceeded by up to 10 hours only while en route to a place where the inspection can be done, and that extra time then counts against the next 100-hour interval.
20. What is an Airworthiness Directive (AD)? **A:** A mandatory FAA-issued correction for a known safety-of-flight issue — functionally a recall. Compliance is mandatory within the AD's specified timeframe or the aircraft is not airworthy.

21. What is a Special Flight Permit (ferry permit), and when would you need one? **A:** An FAA authorization to fly an aircraft that doesn't currently meet airworthiness requirements — e.g., overdue for annual, or needing a repair not available at your location — to a place where the work can be done, or for delivery/export/relocation. Issued by the FSDO with specific conditions and a limited route/duration.
22. Preflight, you find the fuel gauge reading zero even with full tanks. Legal to fly? **A:** No. A fuel quantity indicator is required equipment under 91.205 for both day and night VFR — it can't be deferred.
23. Your turn coordinator isn't working. Can you fly, and if so, what do you have to do first? **A:** Walk the decision tree in 91.213: check for an MEL (Minimum Equipment List — most training aircraft don't have one, so 91.213(d) governs instead) — if none, check if the item is required by 91.205, an AD, or the aircraft's type certificate/KOEL (Kinds of Operation Equipment List, found in the POH limitations section). A turn coordinator is not required for day VFR, so you can fly, but a certificated mechanic must determine it's safe to fly without it, and it must be placarded inoperative and logged so maintenance is aware.
24. What equipment is required for VFR day flight? **A:** Airspeed indicator, Tachometer (per engine), Oil pressure gauge, Manifold pressure gauge (for each altitude engine), Altimeter, Temperature gauge (per liquid-cooled engine), Oil temperature gauge (per air-cooled engine), Fuel gauge (per tank), Landing gear position indicator (retractable gear), Anti-collision lights, Magnetic compass, ELT, and seat belts (plus shoulder harnesses for required crew). (Mnemonic: A TOMATO FLAMES.) [91.205(b)]
25. What additional equipment is required for VFR night flight? **A:** On top of the day list: Fuses (a spare set, or three of each kind), Landing light (only if operated for hire), Anti-collision lights, Position/navigation lights, and a Source of electrical power adequate for the equipment used. (Mnemonic: FLAPS.) [91.205(c)]
26. What kinds of inoperative equipment can a private pilot personally deactivate and placard, versus what requires a mechanic? **A:** Very little — a pilot generally cannot make that determination or disable equipment themselves. If an item isn't required by the type certificate, an AD, 91.205, or the kind of operation being flown, it may be placarded "Inoperative" and deactivated under 91.213(d), but a certificated mechanic must determine it's safe to fly without it first.
27. What preventive maintenance can you legally perform, and on what aircraft? **A:** A Part 61 pilot certificate holder (other than sport pilot, which has its own narrower list) may perform preventive maintenance on any aircraft they own or operate — not just one they own outright — as long as it isn't operated under Part 121, 129, or 135. This applies to a student pilot certificate holder too. The roughly 30 items on the Part 43 Appendix A(c) list — oil/filter changes, servicing tires, replacing bulbs/fuses, spark plug cleaning, and similar — don't require any additional certificate, but the work must be logged to be approved for return to service.
28. When must a maintenance logbook entry be made, and by whom? **A:** Immediately after any inspection, maintenance, preventive maintenance, alteration, or rebuild, and before the aircraft is approved for return to service — describing the work performed and signed by the person who performed it.

3. Aircraft Systems

29. Describe your aircraft's engine. **A:** Give the real specifics: manufacturer and model (e.g., Lycoming O-320), horizontally opposed, air-cooled, normally aspirated or turbocharged, fuel-injected or carbureted, direct-drive to a fixed or constant-speed prop, and horsepower rating. Know your specific airplane's actual numbers, not a generic mnemonic.
30. Is your engine fuel-injected or carbureted, and what's the practical downside of each? **A:** A carbureted engine is simpler and cheaper to maintain but susceptible to carburetor icing even on a clear day with temperatures well above freezing (high humidity is the real trigger). Fuel-injected engines avoid carb ice but are typically more expensive to maintain and can be harder to hot-start.
31. How would you recognize carburetor icing, and is the onset sudden or gradual? **A:** Gradual — a slow, unexplained drop in RPM (fixed-pitch) or manifold pressure (constant-speed), engine roughness, and possibly a rough-running feel before it worsens. It can form even in outside air temperatures well above freezing (roughly 20°F–70°F) with visible moisture or high humidity, because of the venturi and fuel-vaporization cooling effect in the carburetor throat. Apply full carb heat immediately when suspected, expect a further momentary RPM drop as the ice melts and is ingested, followed by a rise as it clears.
32. What keeps the engine running if the battery and electrical system fail completely? **A:** The magnetos — self-contained, engine-driven ignition sources independent of the aircraft's electrical/battery system. Most piston aircraft have two for redundancy, each firing one of two spark plugs per cylinder.
33. How can you tell if one magneto has actually failed, versus just switching between them normally? **A:** Run the engine on "both," then switch to one magneto at a time. A normal, healthy magneto will show a small, expected RPM drop when isolated. If switching to one side causes no change at all from "both," that's actually a sign the *other* magneto wasn't contributing in the first place — a counterintuitive detail worth knowing cold. During runup, an excessive drop (beyond the POH limit, or an excessive left/right difference) usually points to a fouled plug, bad wire, or timing issue.
34. What are the indications of an alternator failure in flight, and what's your response? **A:** An ammeter/load meter reading zero or a lit alternator warning light are the indications. Response: run the checklist — typically verify the failure, cycle the alternator/master switch off then back on (this can reset an over-voltage protection circuit that may have tripped it offline, or clear a tripped alternator field circuit breaker), reduce electrical load to conserve battery, and land as soon as practical since the battery alone only provides limited endurance.
35. If the master switch is turned off entirely, does the engine keep running? **A:** Yes — the engine runs on magneto ignition independent of the electrical/master system; only the electrically-dependent equipment (avionics, lights, electric flaps if equipped) goes dark.
36. What's the difference between best power mixture and best economy (lean-of-peak) mixture, in plain terms? **A:** Leaning reduces fuel flow, which first increases exhaust gas temperature (EGT) to a peak, then decreases it further past peak. Best power is typically set rich of that peak; best economy/lean-of-peak is set on the far side of it. Running very lean-of-peak can save fuel but pushes some engines closer to detonation risk if not done exactly per the manufacturer's specific guidance — which is why some POHs explicitly restrict it.

4. Weather

37. How can you obtain a weather briefing before flight, and what types are there? **A:** 1800wxbrief.com, the ForeFlight/Garmin Pilot app, AviationWeather.gov, or calling Flight Service (1-800-WX-BRIEF). Three briefing types: a Standard briefing (complete, used for the first briefing of the day or after 6+ hours since the last one), an Abbreviated briefing (updates a previous one or covers specific information only), and an Outlook briefing (used when departure is 6+ hours away, for planning purposes only — must be followed by a standard or abbreviated briefing closer to departure).

38. What's a METAR, and what's a TAF? **A:** A METAR is a routine, actual-conditions surface observation issued hourly, or as a special/SPECI report when conditions change significantly. A TAF is a forecast, issued roughly every 6 hours, valid 24–30 hours, describing expected conditions within about 5 statute miles of the airport.

39. What is a PIREP, and why is it valuable? **A:** A pilot report of actual in-flight conditions — turbulence, icing, cloud tops/bases, visibility — relayed to ATC or Flight Service. It's the only "ground truth" source of current en route conditions; forecasts are inferential, a PIREP is an eyewitness account. Pilots are encouraged (and, for icing or wind shear, urged) to file them.

40. What are AIRMETs and SIGMETs, how do they differ, and how long is each valid? **A:** SIGMETs (Significant Meteorological Information) cover hazards significant to all aircraft — severe icing, severe/extreme turbulence, dust/sandstorms below 3 SM visibility, volcanic ash — and are valid up to 4 hours (6 for hurricanes). Convective SIGMETs cover severe thunderstorm activity specifically and are valid 2 hours. AIRMETs cover less severe conditions but of particular concern to lighter aircraft, issued every 6 hours in three types: Sierra (IFR conditions/extensive mountain obscuration), Tango (moderate turbulence, sustained winds ≥ 30 kt, or low-level wind shear), and Zulu (moderate icing and freezing levels).

41. What are the categories of turbulence, and how are they defined? **A:** Light, moderate, severe, and extreme — defined by the effect on the aircraft (momentary bumps vs. large abrupt changes vs. occupants forced against restraints vs. aircraft momentarily out of control), not by a specific wind value. PIREPs are the most reliable source of actual turbulence conditions, since forecasts only outline areas of probability.

42. What are the types of structural icing, and what two conditions does icing require? **A:** Rime ice — rough, milky, opaque, forms quickly from small supercooled droplets, typical in stratiform clouds. Clear ice — smooth, hard, transparent, and the most dangerous/hardest to remove, from larger supercooled droplets that spread before freezing (freezing rain/drizzle, cumuliform clouds). Mixed ice — a rough combination of both. Structural icing always requires two things: visible moisture, and a temperature at or below freezing at the airframe surface (not necessarily the ambient air).

43. What are the VFR weather minimums? **A:** Class B: 3 SM visibility, clear of clouds. Class C/D/E below 10,000 ft MSL: 3 SM, 500 ft below/1,000 ft above/2,000 ft horizontal from clouds. Class E at or above 10,000 ft MSL: 5 SM, 1,000 ft below/1,000 ft above/1 SM horizontal. Class G at or below 1,200 ft AGL: 1 SM day (clear of clouds) / 3 SM night (500/1,000/2,000). Class G above 1,200 ft AGL, below 10,000 ft MSL: 1 SM day / 3 SM night, 500/1,000/2,000. Class G above 1,200 ft AGL, at/above 10,000 ft MSL: 5 SM, 1,000/1,000/1 SM.

44. What are the different types of fog, and how does each form? **A:** Radiation fog forms over land on clear, calm nights as the surface radiates heat away (common in valleys), and usually burns off after sunrise. Advection fog forms when warm, moist air moves over a cooler surface (common along coasts) and can persist until the wind or air mass changes. Upslope fog forms as moist stable air is mechanically lifted up rising terrain and cools to its dew point. Steam fog forms when cold, dry air moves over warm water. Precipitation-induced (frontal) fog forms when warm rain falls through cooler air and raises its dew point to saturation, associated with warm fronts and slow-moving cold fronts.
45. How do high- and low-pressure systems behave in the Northern Hemisphere, and why does that matter for altimeter settings? **A:** High pressure circulates clockwise and outward, generally bringing fair weather; low pressure circulates counterclockwise and inward, generally bringing clouds and precipitation. "Flying from high to low, look out below" — flying toward lower pressure or warmer temperature without resetting the altimeter causes it to read higher than your true altitude, meaning you're actually lower than indicated.
46. Describe a warm front, a cold front, an occluded front, and a stationary front. **A:** Warm front — warm air rides up over retreating cooler air on a shallow slope, producing widespread stratiform clouds and steady precipitation well ahead of the surface front, moving slowly. Cold front — cold, dense air wedges under and lifts warm air abruptly on a steep slope, producing a narrow band of cumuliform clouds, showery/heavy precipitation, turbulence, and sometimes thunderstorms, moving fast with rapid clearing behind it. Occluded front — a faster cold front catches a slower warm front; expect warm-front weather followed by cold-front weather in sequence. Stationary front — two air masses meet with little relative motion, and the associated weather can linger over the same area for days.
47. What three ingredients does a thunderstorm need to form? **A:** Sufficient moisture, an unstable air mass (a lapse rate that lets a rising parcel stay warmer than its surroundings), and a lifting force (frontal, orographic, or surface heating/convection).
48. What does it mean when temperature and dew point converge? **A:** The air is approaching saturation, favoring fog, low clouds, and haze, and increasing the likelihood of IMC. A spread of 2–3°C or less that's narrowing is a classic warning sign on a METAR or TAF.
49. If you can't see the ground because you're above a cloud layer, are you automatically flying illegally? **A:** Not necessarily — this can be legal "VFR over the top" as long as you maintain the applicable cloud clearance and visibility minimums for the airspace you're in. It's distinct from "VFR-on-top," which is an IFR clearance. It's legal but not always wise — you need a way back down through a gap, and rising cloud tops (common on summer afternoons) can trap you above the layer burning fuel.
50. Does density altitude only matter for calculating takeoff and landing distance? **A:** No — it also affects climb performance after you're airborne. A runway can be more than long enough on a hot, high-elevation day while the aircraft still can't outclimb rising terrain ahead, which is a common trap if you only check the takeoff-distance chart and stop there.
51. You inadvertently enter IMC as a VFR-only pilot. Walk through what you do. **A:** First, don't panic — rely on the instruments rather than sensory impressions (which will be unreliable/disorienting). Make a standard-rate 180° turn back toward the VMC you just left (a rule of thumb: bank angle roughly equal to the last two digits dropped from your airspeed in knots — e.g., ~11° at 110 knots). If that doesn't get you back to VMC within a minute or two, contact ATC (or Flight Following if already established), explicitly state you are

a non-instrument-rated pilot inadvertently in IMC, and request vectors to the nearest VFR conditions/airport. If needed, declare an emergency and squawk 7700.

52. What are personal minimums, and how do you set them — what's the PAVE model? **A:** Self-imposed limits stricter than legal/regulatory minimums, set in advance rather than in the moment, to account for your actual experience level, recency, and the mission at hand. PAVE is a tool for setting and reviewing them: Pilot (your own currency/proficiency/fitness), Aircraft (its capability and equipment for the mission), enVironment (weather, terrain, airport conditions), and External pressures (schedule, passengers, get-there-itis). Knowing your minimums is just a number on paper — living by them under real pressure is where most weather-related fatal accidents actually get decided, when a pilot overrides a minimum they already knew.

5. Airspace

53. Name the six classes of airspace and how their requirements differ. **A:** Class A — 18,000 ft MSL up to FL600, IFR only. Class B — around the busiest airports, requires an explicit ATC clearance, two-way radio, Mode C transponder, and ADS-B Out; depicted with solid blue rings. Class C — around busy radar-served airports, requires two-way radio contact established (not necessarily an explicit clearance) before entry, Mode C transponder required; solid magenta rings. Class D — around towered airports without approach control, two-way radio contact required; dashed blue circle. Class E — controlled airspace not otherwise classified, joining G at various floors; magenta or blue shading/lines depending on the floor. Class G — uncontrolled airspace, simply the absence of B/C/D/E markings.

54. What special use and other airspace types should you know? **A:** Prohibited Area — flight prohibited entirely (e.g., Washington, D.C.). Restricted Area — hazardous activity like artillery/gunnery; entry prohibited when active without permission. Warning Area — same hazards as Restricted but over international waters. Military Operations Area (MOA) — military training activity; VFR flight is permitted but caution advised, check for activity. Alert Area — high volume of training or unusual aerial activity; no special clearance required. Controlled Firing Area (CFA) — activity suspends if a spotter detects an inbound aircraft; not depicted since it doesn't affect other traffic. National Security Area (NSA) — pilots requested, not required, to voluntarily avoid. TFR — issued by NOTAM for events like VIP movement, disasters, or firefighting; typically mandatory to avoid.

55. What is a TFR, and how do you find out about one before a flight? **A:** A Temporary Flight Restriction — airspace temporarily closed or restricted, commonly for VIP movement, wildfires, major sporting events, or disaster response. Check NOTAMs during your preflight weather/airspace briefing every time, since a TFR can appear on short notice even along a route you've flown before.

56. What is Special VFR, and when might you use it? **A:** ATC-authorized VFR flight within the surface area of Class B, C, D, or E airspace when weather is below basic VFR minimums but the pilot can remain clear of clouds with at least 1 statute mile visibility during the day. Must be requested by the pilot — ATC can't require it except in an emergency. At night, both the pilot must be instrument-rated and current and the aircraft IFR-equipped.

57. Is flying just below the Class B shelf a good way to avoid dealing with busy airspace and traffic? **A:** Not necessarily — it can actually put you into more congestion, since a lot of other traffic avoids the Bravo the same way, and ATC may not be talking to any of you. Many Bravo areas publish VFR flyway routes (on the back of some Terminal Area Charts) specifically to help deconflict that traffic; a VFR corridor, distinct from a flyway, is a defined tunnel straight through Class B airspace that doesn't require a clearance or ATC contact at all. Both are advisory planning tools, not regulatory shortcuts around the airspace rules.

58. What are the VFR cruising altitude rules? **A:** Above 3,000 ft AGL, in level cruising flight: magnetic course 0°–179° flies odd thousands + 500 ft (3,500, 5,500, 7,500...); magnetic course 180°–359° flies even thousands + 500 ft (4,500, 6,500, 8,500...).

59. What are the transponder emergency codes, and what does "squawk VFR" or an "ident" instruction mean? **A:** 1200 is the standard VFR code for aircraft not receiving ATC services. 7500 means hijacking, 7600 means lost communications, 7700 means a general emergency. "Ident" instructs you to press the transponder's IDENT button, which highlights your specific return on the controller's scope for positive identification.

60. What are the basic right-of-way rules? **A:** An aircraft in distress has right of way over all other traffic. When converging at the same altitude (other than head-on), the aircraft to the other's right has the right of way — except a balloon, glider, or aircraft towing/refueling has right of way over powered aircraft. Head-on: both aircraft alter course to the right. Overtaking: the aircraft being overtaken has the right of way, and the overtaking aircraft passes to the right. On landing or final approach, the aircraft at the lowest altitude has right of way — but must not use that to cut off another aircraft already established closer to the runway.

61. What are NOTAMs, and where do you find them? **A:** Notices to Air Missions — time-critical information not known far enough in advance to be published elsewhere: runway/taxiway closures, NAVAID or lighting outages, temporary obstructions, TFRs, and airspace changes. Found via 1800wxbrief.com, ForeFlight, or a Flight Service briefing; a full weather briefing automatically includes applicable NOTAMs.

62. What is an ADIZ, and where might you encounter one? **A:** An Air Defense Identification Zone — airspace where an aircraft must file a flight plan, maintain two-way radio contact, and have an operating transponder before entry, for national security identification. The Washington, D.C. Special Flight Rules Area is the one most GA pilots in the eastern US need to know.

6. Cross-Country Flight Planning

63. What do you consider when planning a cross-country route? **A:** Airspace you'll transit (and any special-use airspace to avoid or get clearance through), terrain and obstacles, available checkpoints/navigation aids, weather along the route, and suitable alternate/diversion airports in case of a problem.

64. How do you select a cruising altitude for a VFR cross-country? **A:** Balance terrain/obstacle clearance, winds aloft (favorable altitude for tailwind/less headwind), and the VFR hemispheric cruising altitude rule above 3,000 ft AGL.

65. What are the required elements of a VFR flight plan, and are you required to file one? **A:** Aircraft identification/type, true airspeed, departure point and time, cruising altitude, route, destination and estimated time en route, fuel on board, and number of persons on board. Filing one is voluntary within the US (with narrow exceptions like some international/ADIZ crossings) — it exists purely for search-and-rescue protection, since Flight Service starts a search if you don't close it within about 30 minutes of your estimated arrival.
66. How do you activate and close a VFR flight plan? **A:** Activate it with Flight Service once airborne — filing it alone does not activate it. Close it the same way promptly after landing, or the search process needlessly begins.
67. What do you check for along your route besides weather? **A:** NOTAMs for runway/taxiway closures, temporary obstacles, or airspace changes; the Chart Supplement for airport-specific information; and current sectional charts for any airspace or obstacle changes since your last flight in the area.
68. How do you use a sectional chart to plan a route, and what does true course become by the time you actually fly it? **A:** Identify airspace boundaries and floors/ceilings, terrain/obstruction elevations, suitable checkpoints, special use airspace, and available nav aids, then plot the true course line with a plotter. That true course is corrected for magnetic variation to get magnetic course, corrected for wind to get true heading, corrected for variation again to get magnetic heading, then corrected for the aircraft's own compass deviation to get compass heading — the number you actually fly. (Mnemonic: True Virgins Make Dull Companions — True, Variation, Magnetic, Deviation, Compass; and "East is least, west is best" for applying variation.)
69. What fuel reserve is required for a VFR flight? **A:** Enough to fly to the first point of intended landing and, at normal cruise, continue flying for at least 30 minutes during the day or 45 minutes at night.

7. Aerodynamics & Performance

70. What is angle of attack? **A:** The acute angle between the wing's chord line (a straight line from leading edge to trailing edge) and the relative wind — the direction the air is actually meeting the wing.
71. Explain the two competing theories of how a wing generates lift. **A:** Bernoulli's principle — as air meets a restriction like the wing's curved upper surface, it speeds up and pressure drops, creating a pressure differential between the upper and lower surface. Newton's third law — the wing deflects air downward and rearward; the equal-and-opposite reaction pushes the wing up. Both act together to produce lift; it isn't either/or.
72. When does an aircraft stall, and at what airspeed? **A:** A stall occurs whenever the wing exceeds its critical angle of attack — it can happen at any airspeed, any pitch attitude, and any power setting. A published "stall speed" only describes the specific 1G, coordinated, given-weight-and-configuration condition most often demonstrated; exceeding critical AOA is what actually causes the stall, not a number on the airspeed indicator.
73. What is load factor, and how does it relate to stall speed? **A:** The ratio of the load the wings are producing (lift) to the aircraft's actual weight, expressed in G's — a load factor of 2 means the wings are

producing lift equal to twice the airplane's weight. Load factor increases with bank angle in a level turn (2G at 60° bank) and increases stall speed by the square root of the load factor (a 2G load increases stall speed by about 41%).

74. Name the left-turning tendencies present during takeoff and climb, and briefly explain each. **A:** Torque (Newton's third law — the engine spins the prop right, so an equal/opposite reaction tends to roll/yaw the airframe left; most pronounced at low airspeed/high power). P-factor (asymmetric propeller loading — with the nose pitched up, the descending blade takes a bigger "bite" of air than the ascending blade, yawing the nose left; most pronounced at high power, low airspeed, high AOA). Spiraling slipstream (the corkscrewing propwash wraps around the fuselage and strikes the left side of the tail, pushing the nose left; most pronounced at low airspeed/high power). Gyroscopic precession (a force applied to a spinning gyroscope is felt 90° later in the direction of rotation — classically, raising the tail on a tailwheel takeoff roll manifests as a left yaw 90° later; most pronounced when the propeller's plane of rotation is displaced).

75. What is wake turbulence, which aircraft generate the most of it, and how do you avoid it? **A:** Counter-rotating vortices generated off the wingtips of any lift-producing aircraft (including helicopters). The worst wake comes from aircraft that are heavy, clean (no flaps/gear extended), and slow. Avoid it by rotating before, and landing beyond, the point where a larger preceding aircraft rotated or touched down, and staying above and upwind of its flight path.

76. What is ground effect? **A:** A reduction in induced drag (and a slight increase in effective lift) when the wing operates within roughly one wingspan of the ground, because the ground restricts wingtip vortex formation and downwash. It can cause an airplane to lift off before reaching best-rate-of-climb speed and to float during landing if excess airspeed is carried into the flare.

77. What's the difference between indicated, calibrated, true, and groundspeed? **A:** Indicated airspeed (IAS) is what the instrument reads. Calibrated airspeed (CAS) corrects IAS for instrument and position error. True airspeed (TAS) corrects CAS for nonstandard pressure and temperature (density altitude) — TAS increases relative to CAS as altitude increases. Groundspeed is TAS corrected for wind — the actual speed over the ground.

78. What is adverse yaw, and how is it countered? **A:** When rolling into a turn, the down-going aileron on the rising wing produces more lift and more induced drag than the up-going aileron on the lowering wing, yawing the nose toward the outside of the turn. Coordinated rudder pressure in the direction of the turn counteracts it — the reason ailerons and rudder are used together.

79. If you enter spatial disorientation, what's the first instrument you should reference to begin recovery — and why not the attitude indicator? **A:** The airspeed indicator. The attitude indicator is a gyroscopic instrument that can tumble and give an unreliable reading once bank exceeds roughly 60° (on older vacuum instruments) — exactly the situation you're in during disorientation. Airspeed trending fast means the nose is likely low (reduce power, raise the nose); airspeed trending toward a stall warning means the nose is likely high (lower the nose, add power). Stabilize on airspeed first, then cross-check the other instruments.

80. What's VA (maneuvering speed), what does it actually protect you from, and why does it change with weight? **A:** The maximum speed at which a single, full, abrupt control deflection will stall the wing before exceeding the airplane's structural load limit — it protects the airframe by ensuring the wing stalls (shedding the load) before the structure fails. It does *not* protect against multiple simultaneous full-deflection inputs

(e.g., full aileron then immediately full opposite rudder), which can still overstress the aircraft even below VA. VA decreases as weight decreases, because a lighter airplane stalls at a lower speed and reaches critical AOA — shedding the excess load — sooner.

81. What's the practical difference between VX and VY, and how do they change with altitude? **A:** VX (best angle of climb) gains the most altitude over the shortest horizontal distance — used to clear an obstacle close in. VY (best rate of climb) gains the most altitude per unit of time — used for a normal climb to cruise. VX increases and VY decreases with altitude until they converge at the airplane's absolute ceiling.

8. Slow Flight, Stalls & Spins

82. What factors affect the airspeed at which a wing stalls? **A:** Angle of attack is the only direct cause of a stall (exceeding the critical AOA), but weight, load factor (bank angle/G-loading), center of gravity, and power setting all change the airspeed at which that critical AOA is reached in a given configuration.

83. What's the difference in setup between a power-off stall and a power-on stall, and what real-world scenario does each represent? **A:** Power-off stall simulates the approach-to-landing configuration (idle power, flaps/gear as appropriate, descending) — the classic base-to-final accident scenario. Power-on stall simulates a departure/climb configuration (higher power, nose-high) — the classic short-field or obstacle-clearance takeoff scenario, with a more abrupt break and stronger yaw from P-factor.

84. What's the proper stall recovery procedure? **A:** At the first indication: reduce angle of attack (release back-pressure), add maximum available power smoothly, level the wings with coordinated aileron and rudder, and minimize altitude loss. Don't try to pick up a dropped wing with aileron alone — use rudder, or aileron alone can deepen the stall on that wing.

85. What's a secondary stall, and what's an accelerated stall? **A:** A secondary stall happens when you pitch up too abruptly during recovery, re-exceeding critical AOA before regaining airspeed. An accelerated stall happens at higher-than-normal airspeed due to increased load factor — a steep turn, for example — a reminder that critical AOA, not a specific airspeed, is always what actually triggers a stall.

86. Why is a cross-control stall especially dangerous? **A:** Because it's typically uncoordinated — rudder and aileron working against each other while overshooting the turn from base to final — which causes one wing to stall before the other. That asymmetric stall can snap the aircraft into a spin at pattern altitude, too low to recover before ground contact.

87. What causes a spin, and how does it differ from a spiral dive? **A:** A spin always begins with a stall — specifically an uncoordinated stall where one wing is stalled more than the other, causing autorotation around a mostly-vertical axis internal to the aircraft. A spiral is not stalled at all — it's a steepening diving turn rotating around an external point, with airspeed increasing rather than staying near stall speed.

88. What's the standard spin recovery procedure, and why does it need to be automatic rather than looked up? **A:** PARE — Power to idle, Ailerons neutral, Rudder full opposite the spin direction, Elevator briskly forward to break the stall — then neutralize rudder and recover from the resulting dive (always follow your specific POH's procedure if it differs). It has to be immediate and reflexive because the most common place an inadvertent spin actually happens is in the traffic pattern — classically, skidding the base-to-final turn to

avoid overshooting centerline — at an altitude that leaves no time to consult a checklist.

9. Human Factors & Aeronautical Decision-Making

89. What is hypoxia — give the types, symptoms, and corrective action. **A:** Insufficient oxygen to body tissue, particularly the brain. Four types: hypoxic (not enough oxygen in the air — altitude), hypemic (blood can't carry oxygen — anemia, CO poisoning, blood donation), stagnant (poor circulation — cold, high G-forces), and histotoxic (cells can't use the oxygen they have — alcohol or drugs). Symptoms: euphoria, poor judgment, headache, drowsiness, tunnel vision, bluish lips/nail beds, and eventually unconsciousness — often without the pilot recognizing the impairment. Corrective action: descend, use supplemental oxygen if available, treat the underlying cause.

90. At what altitudes is supplemental oxygen required? **A:** Required crew must use it above 12,500 ft MSL for that portion of flight lasting more than 30 minutes, and continuously above 14,000 ft MSL. Above 15,000 ft MSL, every occupant must be provided (not just offered) supplemental oxygen.

91. A passenger is hyperventilating and can't get their breathing under control. What do you do — and why not hand them a paper bag? **A:** Have them read a checklist aloud or talk through something continuously — this naturally slows and regulates breathing, since hyperventilation is abnormally rapid/deep breathing that washes out CO₂ faster than the body produces it. Avoid the classic "paper bag" fix: at typical cruising altitudes, rebreathing exhaled air can tip someone from hyperventilating straight into hypoxia, which is worse and harder to notice.

92. What causes ear and sinus problems in flight, and how do you prevent or correct them? **A:** Pain from the eustachian tubes being unable to equalize pressure between the middle ear/sinuses and the cabin, usually during descent and worsened by congestion from a cold or allergies. Correct it with the Valsalva maneuver (pinch nose, gently blow), yawning, or swallowing; the best prevention is not flying with an upper respiratory infection or bad allergies in the first place.

93. What is carbon monoxide poisoning, and how do you guard against it? **A:** CO is a colorless, odorless gas from engine exhaust that can leak into the cabin — commonly through a cracked heat exchanger feeding the cabin heater — and binds to hemoglobin far more readily than oxygen, causing a form of hypemic hypoxia. Symptoms (headache, dizziness, confusion, drowsiness) are easily mistaken for ordinary fatigue. Response: cabin heat off, fresh air vents open, land as soon as practical; carry and heed a CO detector.

94. What optical illusions should a pilot be aware of? **A:** Runway width/slope illusions (a narrower-than-normal runway makes you feel too high, a wider one too low; an upsloping runway makes you feel too high, downsloping too low). Featureless terrain illusion causes flying a lower-than-normal approach. Black hole approach — over dark terrain or water with few lights, the runway appears farther away or higher than it actually is, especially dangerous with no visible horizon. Autokinesis — a stationary light at night appears to move after being stared at. All are countered by cross-checking instruments and known glidepath references (VASI/PAPI) rather than trusting visual impression alone.

95. How long should you wait to fly after scuba diving? **A:** At least 12 hours after a dive that didn't require a controlled ascent (no-decompression dive); at least 24 hours after any dive requiring a decompression stop, or after multiple/repetitive dives. Flying too soon risks decompression sickness as reduced cabin pressure lets dissolved nitrogen form bubbles in the bloodstream.
96. What are the hazardous attitudes, and what's the antidote to each? **A:** Anti-authority ("don't tell me") — antidote: follow the rules, they're usually right. Impulsivity ("do it now") — antidote: not so fast, think first. Invulnerability ("it won't happen to me") — antidote: it could happen to me. Macho ("I can do it") — antidote: taking chances is foolish. Resignation ("what's the use") — antidote: I'm not helpless, I can make a difference.
97. What is Aeronautical Decision-Making (ADM), and what are SRM and CRM? **A:** ADM is a systematic, continuous process for identifying hazards, assessing risk, and choosing the best course of action given the time and resources available. CRM (Crew Resource Management) is the use of all available resources by a multi-pilot crew; SRM (Single-Pilot Resource Management) is the same concept applied by one pilot managing all internal (aircraft systems, personal condition) and external (ATC, weather, passengers) resources alone.
98. What's the difference between knowing your personal minimums and actually living by them? **A:** Knowing them is just a number on paper. Living by them means holding to that number under real pressure — a passenger's schedule, a bill that needs the flight to happen, a "get-there-itis" mindset — which is where most weather-related fatal accidents actually originate: not from ignorance of the risk, but from a pilot overriding a minimum they already knew.
99. What are the three possible outcomes of a checkride, and what specifically distinguishes "incomplete" from "unsatisfactory"? **A:** Satisfactory (consistent performance within ACS standards — not flawless, but correcting deviations appropriately), Unsatisfactory (consistently outside standards, or an action that jeopardizes safety), and Incomplete (the test can't be finished for a reason outside the applicant's knowledge/skill — weather, an aircraft airworthiness issue, sudden illness). Being unprepared, or simply not knowing an answer, is never grounds for "incomplete" — that's the applicant's own responsibility. [14 CFR 61.43(b)]

10. Emergency Procedures

100. What would you do for a partial power loss (rough-running engine)? **A:** Apply carburetor heat first — the most common cause of partial power loss is carb ice. Check and adjust mixture, check magnetos individually (a fouled plug may run better on one), check the fuel selector and switch tanks if warranted, and scan engine gauges (oil pressure/temp, fuel pressure) for a diagnosable cause. If power is sufficient to maintain flight, divert to the nearest suitable airport; if it keeps degrading, treat it as an impending complete failure and prepare for a forced landing.
101. Engine fails at 200 feet right after takeoff. What do you do? **A:** Land straight ahead (with minor turns only to avoid an obstacle), pitch for best glide immediately. At that altitude there is essentially never enough height to turn back to the runway — attempting the "impossible turn" is one of the classic fatal-accident scenarios in GA. Brief the plan before every takeoff: a pre-briefed altitude below which you land straight

ahead (roughly $\pm 30^\circ$), and only consider turning back if you've practiced it and are above that altitude.

102. Engine fails at cruise altitude with plenty of altitude to work with. Walk through your priorities. **A:** Fly the airplane first — pitch immediately for best glide speed and pick a suitable landing area, considering wind, terrain, and obstacles. Only after that's established do you run the engine-restart checklist (fuel selector/tank, mixture, carb heat/fuel pump, primer, mags) — altitude is time, so use it to troubleshoot, but never let troubleshooting come before controlling the aircraft and picking a spot. If restart fails, secure the engine if time allows, declare an emergency if time/altitude allow, and brief passengers before configuring for landing.

103. What's the difference between best glide speed and minimum sink speed, and why does it matter which one you use? **A:** Best glide gives you the greatest horizontal distance for a given altitude loss — what you want if you need to reach a specific landing spot. Minimum sink gives you the longest time aloft (but not the most distance) — more relevant if you're trying to buy time, e.g. for troubleshooting, rather than stretch a glide to a specific point.

104. How do you choose a forced-landing site, and what pattern do you fly to it? **A:** Prioritize a site that's long, into the wind if practical, free of obstacles, and reachable without maneuvering below pattern altitude too early — a firm, open field beats a marginal attempt to stretch the glide to an airport. Fly a modified pattern (often a "key position" abeam your intended touchdown point at normal pattern altitude/distance) to manage energy and arrive with options, using flaps or slips to adjust the glidepath, and plan to touch down at minimum controllable airspeed.

105. You get an engine fire during engine start on the ground. What's your procedure? **A:** Continue cranking (to try to draw the fire back into the engine and starve it), mixture to idle cutoff, throttle full open, fuel selector off. If the fire continues after that, evacuate the aircraft and use an extinguisher if available/safe.

106. How does an in-flight engine fire response differ from an electrical fire? **A:** For an engine fire: mixture to idle cutoff, fuel selector/pump off to starve it, cabin heat and air off (to avoid drawing smoke into the cabin through the heater muff), and turn the aircraft so the fire-side wing is into the wind to help blow it out, before transitioning to best glide and a forced landing. Don't attempt a restart once the engine is secured. For an electrical fire: master switch off (killing the electrical source) first, then address individual equipment — you don't want to shut off fuel to an engine fire that doesn't exist.

107. What would you do if a door popped open in flight? **A:** Recognize it as a distraction, not an emergency — the airplane will fly essentially normally with a door open, just louder and draggier. Maintain aircraft control first, then attempt to close it only if it can be done safely (usually by slowing down and opening a window/vent to equalize pressure); if it won't close, land as a precaution rather than fixating on it in flight.

108. What's your responsibility, and ATC's role, if you declare an emergency? **A:** As PIC, in an emergency you have the authority to deviate from any regulation to the extent required to meet that emergency, though you may be asked to file a report afterward. ATC can provide priority handling, vectors, and coordination with emergency services on the ground — declaring early gives them more time to help, so there's little downside to declaring sooner rather than later if you're unsure. [91.3(b)]

109. What are your notification and reporting responsibilities after an accident or serious incident? **A:** Notify the NTSB "immediately" after an accident, or after specific serious incidents like a flight control malfunction

or in-flight fire, and file a written report within 10 days of an accident (7 days for an overdue-aircraft report). Preserve the wreckage and all records until the NTSB releases them, except as needed to protect life/property or in cooperation with law enforcement. [49 CFR Part 830]

110. What emergency equipment is required, and what should you carry beyond the minimum? **A:**

Required (with limited exceptions): an ELT. A fire extinguisher and first aid kit are commonly required or strongly recommended depending on the operation, even when not federally mandated. Beyond the legal minimum, consider a handheld radio and portable GPS as backups, a flashlight with a red lens for night, water and basic first aid supplies, and for remote/mountainous/wilderness terrain, survival gear suited to the environment and season.

111. How does an ELT work, and how do you check that it's functioning? **A:** It automatically activates on sudden deceleration (a crash impact) and transmits on 121.5/406 MHz to aid search and rescue; a 406 MHz ELT also sends a digital, aircraft-identifying signal to the COSPAS-SARSAT satellite system for a faster, more accurate fix. Operational check: within the first 5 minutes after the hour, monitor for no more than 3 audible sweeps — any longer check should be coordinated with ATC/Flight Service to avoid triggering a false alert.

11. Airport Operations & Communications

112. What ATC light signals might you receive if you lose radio communication in the pattern at a towered airport? **A:** Steady green means cleared for takeoff on the ground, cleared to land in flight. Flashing green means cleared to taxi on the ground, return for landing in flight. Steady red means stop on the ground, give way/continue circling in flight. Flashing red means taxi clear of the runway on the ground, airport unsafe/do not land in flight. Alternating red/green means exercise extreme caution in either case. Know the full table cold — it's the actual fallback for lost comm at a towered field.

113. What's the standard procedure if you lose two-way radio communication at a towered airport while inbound to land? **A:** Squawk 7600, remain outside or above the Class D/C/B surface area until the flow of traffic is determined, then join the pattern and watch for light signals. Continue to comply with the last clearance received. Land when a green light is received or after normal traffic flow makes a safe landing clearly possible.

114. What happens to a Class D airport's airspace when the tower closes for the night? **A:** It typically reverts to Class E (if it has surface-based weather reporting) or Class G, per times published in the chart supplement — no ATC communication is required while the tower is closed, and normal nontowered self-announce procedures apply.

115. At a nontowered airport, how do you determine which traffic pattern and runway to use, and what's the recommended pattern entry? **A:** Check the segmented circle and wind sock/tee at the field for the active runway and pattern direction, and the Chart Supplement for the standard pattern altitude/direction (left traffic unless otherwise noted). The recommended entry is 45° to the downwind leg, at pattern altitude, entering at the midpoint of the downwind.

116. Two aircraft with no radio contact converge to land at the same nontowered airport. Who has right of way? **A:** The aircraft at the lower altitude has right of way, but that aircraft may not use that right to cut off another aircraft on final approach or to overtake it — right-of-way rules never grant permission to be careless. An aircraft already established on final generally has priority regardless of how it got there.

117. Describe the runway lighting and marking systems you might encounter. **A:** Runway edge lights are white (yellow for the last 2,000 ft on instrument runways). Threshold lights are green viewed on approach, red viewed from the runway. REILs are synchronized flashing white lights marking the threshold. A VASI (2-bar) shows red over white for on-glidepath, white over white for too high, red over red for too low ("red over white, you're all right"). A PAPI (4 lights in a row) shows 2 red/2 white for on-glidepath, more white for high, more red for low.

12. Navigation

118. What is the difference between pilotage and dead reckoning? **A:** Pilotage is navigating by visually matching ground landmarks to the chart. Dead reckoning is navigating by calculation — heading, groundspeed, and time derived from true course, wind correction, and true airspeed — without reference to landmarks. VFR cross-country navigation normally combines both: dead reckoning sets up the plan, pilotage confirms and corrects it in real time.

119. What magnetic compass errors should you understand? **A:** Variation is the fixed angular difference between true and magnetic north at a given location. Deviation is compass card error caused by the aircraft's own magnetic/electrical fields, varying by heading and unique to each aircraft (posted on its compass correction card). Dip errors occur during maneuvering: on north/south headings, the compass leads or lags during turns (in the Northern Hemisphere, "accelerate north, decelerate south"); on east/west headings, acceleration or deceleration alone makes the compass show a turn that isn't happening.

120. How does VOR navigation work, and how often must a VOR receiver be checked? **A:** A VOR ground station lets an aircraft receiver determine its magnetic bearing (radial) from the station. Tune and identify the station, set the desired course on the OBS, and read the CDI — centered with a "TO" or "FROM" flag shows you're on that radial's course to or from the station, and each dot of deflection represents about 2° off course. A VOR check is only required if it'll be used for IFR navigation — using a VOT signal, a certified checkpoint, or comparing two receivers against each other — logged within the preceding 30 days.

121. What are the advantages and limitations of GPS versus VOR, and what is RAIM? **A:** GPS gives continuous, accurate position, ground track, and direct point-to-point navigation anywhere, independent of ground infrastructure, but depends on satellite signal integrity and database currency. VOR is a simpler, independent, ground-based backup that doesn't rely on GPS, but only defines specific radials from fixed stations and takes more workload to use for direct navigation. RAIM (Receiver Autonomous Integrity Monitoring) is a GPS receiver's internal check that enough satellites are in view and well-positioned to detect a faulty signal — a RAIM warning means the GPS position can't be trusted and you should cross-check with other means.

122. What radar services are available to VFR pilots? **A:** VFR flight following — traffic advisories from ATC on a workload-permitting basis, valuable but not required — plus radar-based lost-procedures assistance,

vectors around weather, and Class B/C traffic sequencing.

123. How do you divert to an alternate airport in flight? **A:** Identify a suitable alternate on the chart (fuel, runway suitability, weather), turn toward it promptly, and estimate heading and distance using the chart or a simple rule of thumb rather than doing exact calculations while also flying the airplane. Update ATC or Flight Following if in contact.

124. What do you do if you become lost or disoriented in flight? **A:** The "5 C's": Climb (for better visual range and radio/GPS reception), Communicate (contact ATC or Flight Service — 121.5 MHz is monitored widely as the emergency frequency), Confess you're unsure of position (ATC can help via radar — don't be embarrassed to ask), Comply with any instructions given, and Conserve fuel (reduce power to best-range settings while sorting it out). Maintain your heading rather than wandering further, and look for large, distinctive landmarks to correlate with the chart.

13. Basic Instrument Knowledge

125. Why does a private pilot need any instrument flying skill at all? **A:** Part 61 requires 3 hours of instrument training specifically so a private pilot can control the aircraft solely by reference to instruments long enough to escape inadvertent IMC — a survival skill, not a substitute for an instrument rating.

126. What is the basic instrument scan? **A:** A continuous cross-check of the attitude indicator, airspeed, altimeter, vertical speed, heading indicator, and turn coordinator, rather than fixating on any one. Each instrument is "primary" for one parameter (pitch, bank, or power) and supports interpretation of the others.

127. How do you recognize and recover from an unusual attitude by instruments alone? **A:** Nose-low, increasing airspeed: reduce power, level the wings, smoothly raise the nose. Nose-high, decreasing airspeed: add full power, level the wings, smoothly lower the nose. Correct bank and pitch together, reference the attitude indicator, and avoid abrupt control inputs either way.

14. Night Operations

128. How does human vision change at night, and what techniques compensate? **A:** The eye shifts from cone-based (color, sharp central) vision to rod-based (monochrome, peripheral) vision, which takes about 30 minutes to fully adapt and is easily disrupted by bright light exposure. Since rods are concentrated away from the center of the retina, off-center viewing — looking slightly to the side of an object rather than straight at it — is more effective at night. Use dim red or white cockpit lighting to preserve dark adaptation, and avoid bright lights (phone screens, white flashlights) before and during flight.

129. What are the most significant night-specific illusions? **A:** Black hole approach — approaching a runway with little lighting around it, over water or dark terrain, can make the runway appear farther away or higher than it is, leading to a dangerously low approach, especially with no visible horizon. Autokinesis — a stationary light stared at in the dark appears to move. The daytime runway-width/slope and featureless-terrain illusions are also worse at night due to fewer overall visual references. Mitigate by flying a stabilized, instrument-cross-checked approach and trusting a VASI/PAPI over visual impression.

130. What are the night currency and equipment requirements? **A:** Currency: 3 takeoffs and landings to a full stop within the preceding 90 days, done between one hour after sunset and one hour before sunrise, to carry passengers at night. Equipment: position/navigation lights and anti-collision lights from sunset to sunrise, plus the day-VFR equipment list, a landing light if operated for hire, and an adequate electrical source for the equipment in use.

131. What special preflight planning considerations apply to a night cross-country? **A:** Verify all aircraft lights work, carry a flashlight with spare batteries, plan the route over lit/populated areas near airports where possible rather than remote/unlit terrain (to preserve forced-landing options), review terrain and obstruction elevations more conservatively, identify airports along the route with pilot-controlled lighting and their frequencies, and check NOTAMs for lighting outages at the departure/destination/alternate.

132. What additional risk does an engine failure present at night, and how do you mitigate it? **A:** Choosing and judging a landing site is far harder with limited visual cues — terrain, obstacles, and surface condition may not be visible until very late. Fly toward lit areas or known airports when possible, use the landing light only once committed to the approach (to preserve night vision and battery), maintain best-glide speed, and be prepared to land primarily by pitch/airspeed control rather than trying to visually pick a perfect spot.

15. Postflight Procedures

133. What do you do after landing and clearing the runway? **A:** Complete the after-landing checklist only once clear of the runway, behind the hold-short line — never while still on the runway, to keep attention on taxiing safely and avoid a runway incursion. Taxi to parking following ground instructions/markings, then complete the engine shutdown and securing checklist.

134. What must be logged after a flight, and what's the pilot's postflight responsibility? **A:** Log flight time, landings (including night landings, for currency), and any discrepancies noted in flight for maintenance follow-up. The PIC remains responsible for leaving the aircraft in a safe, secure condition — tied down or chocked, controls locked, windows/doors closed — appropriate to the parking location and expected weather.

135. What should you do if you notice new damage or a discrepancy while securing the aircraft? **A:** Note it in the aircraft's discrepancy log or notify the owner/flight school/maintenance directly, and if it affects airworthiness, make sure the aircraft isn't flown again until a mechanic evaluates it — the same PIC responsibility for airworthiness applies at the end of a flight as at the beginning.